

# Transformation by Design: How Decentralisation Rebuilt Texas Instruments, 1985–1996

**Central claim:** Texas Instruments' resurgence from 1985 to 1996 was driven by its strategic transformation of organisational structure to a decentralised model, which redefined its identity and enhanced its adaptability in response to a changing environment — and lifted its stock price from \$1.00 to \$9.00.

## Introduction

---

Texas Instruments operated in a highly competitive semiconductor industry during the 1980s, a period marked by rapid technological advancement and the aggressive entry of Japanese competitors. Teece (2007) emphasises that for an organisation to thrive in such dynamic environments, it must develop and leverage unique dynamic capabilities.

Drawing on Teece's (2007) dynamic capabilities framework, this article explores how TI restructured its organisation to enhance adaptability, improve communication and foster ambidexterity, enabling the firm to regain its competitive edge. TI's segments at the time were Components, Defence Electronics, Digital Products and Metallurgical Materials. Under the leadership of Jerry Junkins, the company reconfigured its management structure, adopting a decentralised model that allowed for more reconfiguring of resources and assets — a shift that positively affected long-term performance.

## Background

---

### A short history of Texas Instruments

Texas Instruments was founded in 1930 as Geophysical Service Inc. (GSI), focusing on oil exploration. By the late 1930s, GSI had diversified into electronics, leveraging its seismic expertise to develop devices for military use during the Second World War, including radar systems and other technologies (Pederson, 2017). This shift marked the beginning of TI's transformation into an electronics powerhouse, and the company adopted the name Texas Instruments to reflect its expansion beyond geophysics. Innovation quickly became its hallmark; the company today carries approximately 80,000 products.

| Year | Milestone                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------|
| 1930 | Geophysical Service Inc. (GSI) founded by Karcher and McDermott                                             |
| 1941 | McDermott, Jonsson and two GSI employees purchase GSI from Coronado                                         |
| 1942 | GSI secures military contracts with the U.S. Navy and Army Signal Corps                                     |
| 1946 | Patrick E. Haggerty joins; the Laboratory and Manufacturing division is created                             |
| 1951 | The L&M division is renamed Texas Instruments Inc.                                                          |
| 1954 | TI and Industrial Engineering Associates develop the world's first small portable radio                     |
| 1958 | Kilby creates the first integrated circuit                                                                  |
| 1961 | TI builds the first computer to use silicon integrated circuits, for the Air Force                          |
| 1967 | Company engineers invent the hand-held calculator                                                           |
| 1970 | The single-chip microprocessor is developed                                                                 |
| 1978 | Speak & Spell launches, using TI's new speech-synthesis technology                                          |
| 1983 | TI posts its first-ever loss, of \$145 million                                                              |
| 1988 | Partnership formed with Hitachi Ltd. to develop 16-megabit DRAM technology                                  |
| 1991 | TI joins Canon, Hewlett-Packard and the Singapore government to build a semiconductor facility in Singapore |

Source: Pederson (2017).

## The strategic situation

Dynamic capabilities refer to an organisation's ability to sense opportunities and threats in the market, seize these opportunities through decisive action, and reconfigure its resources, structures and processes to adapt to changing environments, thereby sustaining a competitive advantage in dynamic contexts (Teece, 2007).

Texas Instruments was a successful firm and the largest producer of semiconductors in the United States. With new entrants, the market environment became more dynamic and more demanding in terms of the cognitive capabilities required of the top management team (TMT). Cognitive capabilities refer to the mental processes and skills that enable individuals and organisations to perceive, interpret and respond to their environments effectively (Neck & Neck, 2022).

The problem was visible to outsiders well before it became a crisis. An article in *The New York Times* quoted Thomas H. Mack, a semiconductor industry analyst at Paine Webber, Mitchell, Hutchins, explaining: "Texas Instruments is a structured, systematic organization, but it often lacks the insight to react to fast-changing market conditions. They run well in a straight line, but they are not so good on the curves" (Schuyten, 1979). During the 1980s, TI's market share in semiconductors fell from 30% to 5% (Burrows, 1993).

*"For decades, Texas Instruments dominated the world semiconductor industry because of its overwhelming strength in technology and manufacturing. But that success caused the company's leaders to believe that the company could do no wrong, according to longtime employees. 'We Got Too Arrogant!'" (Hayes, 1987)*

The TMT at Texas Instruments had employed a resource-based strategy and failed to update its mental models, which led it to believe that high R&D investment and product diversification would deliver success as they had in the past (Teece & Pisano, 1998). Mental models are "empirical representations of an individual's or group's internally held understanding of their external world" (Hoffman et al., 2014). The model formed by the TMT was essentially a simplified understanding of the environment and of the strategy employed by TI.

Analogical thinking allowed the TMT to base its strategy on the company's past success. Analogical thinking is a cognitive process that involves transferring knowledge or insights from one domain (the source) to another (the target) based on perceived similarities between the two (Blanchette & Dunbar, 2000). The previous model had worked with high R&D investment and the exploration of new opportunities to support other business functions. This causal mental model could have led the firm to failure had a new CEO not been appointed.

The period saw significant challenges from Japanese competitors, who disrupted the market with wider technological capabilities and more aggressive pricing strategies (Langlois & Steinmueller, 2000). Texas Instruments was struggling to follow the dynamic nature of the market. Jerry R. Junkins was appointed Chief Executive Officer in 1985, and the transformation of TI began that year when he called his top 20 deputies to Austin, Texas for a weeklong strategic planning exercise which resulted in TI 2000 (Burrows, 1993). Junkins aimed for a decentralised structure, higher adaptability and better communication channels within the firm, with the goal of increasing the organisation's dynamic capabilities. This exercise resulted in the structural transformation that gave Texas Instruments back its competitive edge.

## Cognitive inertia

---

Upper Echelons theory argues that organisational outcomes, including strategies and effectiveness, mirror the values and cognitive frameworks of the TMT within the organisation (Hambrick & Mason, 1984). Managers at TI were anchored in a resource-based strategy, overly focused on technological dominance and high R&D investment, both of which had driven their past successes. This had become part of TI's identity. They engaged in analogical thinking to assess their environment and their mental models, and those models became deeply entrenched, fostering a belief that prior methods would yield future results despite environmental shifts.

Top management viewed the semiconductor market through the lens of its historical strengths, assuming product diversification and innovation would maintain their competitive edge (Langlois & Steinmueller, 2000). This caused them to overlook dynamic market conditions, particularly the

competitive threat posed by Japanese firms with superior pricing and manufacturing strategies. "By contrast, the American firms insisted on radical new designs and process technology, which increased development times and startup problems" (Borrus, 1988).

TI's myopic tendencies were exacerbated by the homogeneity of its top management team and by its reliance on familiar sources of information for decision making. Myopic tendencies can be described as the tendency of groups — in this case the management team — to look for information in familiar places. Research suggests that more homogeneous TMTs are prone to limited and shallow information searches, often confining their focus to familiar territories (Smolinski et al., 2020).

The leadership remained focused on leveraging historical strengths rather than adapting to the competitive advances of Japanese firms and improving processes and specialisation in segments. This prevented effective sensing within the firm and resulted in arrogance and over-confidence on the part of the TMT. "We were so successful in the 1950s and 1960s that we became a real arrogant bunch of young SOBs," admits former CEO Shepherd (Burrows, 1993). The consequence was managers failing to update their business models, and an organisational identity becoming inert.

## Transformation by design

---

The company's own annual reports record the structural transformation as it happened. "TI implemented a new management structure aimed at enhancing its focus on strategic alliances" (Anon., 1986; Anon., 1990). "TI has been engaged, since 1985, in a strategic review process to better position its businesses in areas representing solid, long-term opportunity. The results of this process are beginning to take effect" (Anon., 1987).

Transformation, or reconfiguring, is the last dynamic capability in Teece's framework. It concerns the restructuring and realignment of the firm and its assets (Teece, 2007). TI achieved this level of transformation with the assignment of Junkins as CEO and with the changes made to the structure of the firm. As *The New York Times* reported, "the military unit has produced something else important to Texas Instruments: most of its new top management, including Mr. Junkins. Under his leadership, the culture of Texas Instruments has been undergoing a badly needed transformation" (Hayes, 1987).

Texas Instruments changed the management team first, and afterwards the CEO changed the structure of the organisation. Junkins aimed to prepare the organisation for change and adaptiveness rather than to transform the organisation according to foresight. This dynamic capability of reconfiguring would allow the firm to succeed in the long term by increasing the flexibility and adaptability of the organisation, giving it a competitive edge against the Japanese competition.

## **Decentralisation**

Structure can be defined as the methods by which an organisation divides its work into specific tasks and coordinates them effectively (Mintzberg, 1979). Transformation is crucial for an organisation, and especially in dynamic environments it is important for the organisation to have flexibility and adaptability. Structural changes at Texas Instruments in the 1980s focused on breaking the company into smaller, more manageable segments to enhance agility and responsiveness to a rapidly changing industry environment.

Junkins implemented smaller groups and chains of command in the organisation for higher adaptability in the dynamic environment the firm was operating in (Rosson, 1988). Change in the people (TMT) and the simultaneous change in the structure can lead to higher revenues in dynamic markets (Miller & Chen, 1994).

The transition under Junkins demonstrated the critical interplay between leadership change and structural transformation. It aimed not only at operational improvements but also at fostering a culture of constructive conflict and diversity of thought. By creating smaller, focused units, TI enabled a more entrepreneurial approach within the organisation, encouraging employees at all levels to contribute to strategy and innovation and essentially changing the identity of the organisation. Junkins allowed the managers of the reorganised segments considerable autonomy, enabling them to independently explore new strategies and update their mental models. This approach fostered innovation and adaptability within the organisation, as managers were empowered to make decisions tailored to their specific operational contexts without excessive oversight or constraints (Haleblian & Finkelstein, 1993). By decentralising and empowering smaller groups, TI positioned itself to better manage uncertainty and maintain competitiveness (Langlois & Steinmueller, 2000).

## **Exploration versus exploitation**

By focusing specifically on a limited range of products, Japanese firms successfully challenged American dominance in the 1980s. They leveraged their unique strengths in high-quality manufacturing and capitalised on the advantages provided by their industrial structure, which was well suited for capital-intensive mass production (Langlois & Steinmueller, 2000). By contrast, at the start of the decade American firms such as Texas Instruments focused more on exploration, as this was a more favourable strategy given TI's organisational structure at the time.

By dividing the company into smaller, decentralised units, TI created an environment conducive to contextual ambidexterity, where both exploration and exploitation could coexist more effectively (Ossenbrink et al., 2019). Ambidexterity, in the context of organisational theory, refers to the ability of an organisation to simultaneously pursue both exploration and exploitation activities. Contextual ambidexterity — in which individuals alternate their logic between exploration and exploitation — becomes vital when the environment presents both numerous and uncertain opportunities, compelling organisations to harness the diverse attention and expertise of their frontline employees to effectively navigate complexity and ambiguity (Ossenbrink et al.,

2019). The TMT aimed to achieve ambidexterity through the restructuring of the organisation, and prioritised the creation of shared goals.

"The direct involvement of the CEO and the other top executives sends a clear message to everyone that the success of the team initiative is of paramount importance to the organization" (Rosson, 1988). An interview with a team member after the restructuring reflects that the top management team was heavily involved with the transformation and with the start of new initiatives for specialisation in areas: "Management supported us, they really did. We could come to them with anything, and they were there for us" (Rosson, 1988). Management at TI was able to balance exploitation and exploration contextually through the independence of segments, which allowed flexibility in approach for each segment depending on the optimal strategy.

## Identity dynamics

---

Identity and strategy at Texas Instruments influenced each other, with decisions made by the management team grounded in the organisation's innovative identity. Identification is a powerful tool used by organisations to facilitate change and help employees deal with it. As *The New York Times* observed, "under his leadership, the culture and identity of Texas Instruments has been undergoing a badly needed transformation" (Hayes, 1987).

This transformation emphasised the increased importance of teamwork and organisational innovation alongside the separation of segments. While different segments might adopt varied strategic approaches, the organisation maintained shared objectives and a unified identity characterised by innovative processes, adaptability, and a strong focus on collaboration and belonging within Texas Instruments (Hayes, 1987).

Resistance to change is a common challenge when introducing structural transformations, as it disrupts established routines and challenges long-held norms and practices. At Texas Instruments, this resistance from managers was particularly pronounced because the new structure deviated significantly from the rigid, centralised system they had adhered to for the past decade. The shift to smaller, more agile teams required not only operational changes but also a profound shift in mindset.

According to Rosson (1988), the best solution for resistance to change involves offering strong management to the teams. Once the organisation had an established core strategy and set of goals, employees adapted to the new identity of the organisation. After this adaptation, the identity of the organisation as a dynamic, ever-evolving and ambidextrous one may become a source of resilience in the future (Dick et al., 2018). This identity transformation also has implications for sustaining ambidexterity in the long term.

An organisation that views itself as inherently dynamic is better equipped to balance exploration with exploitation. For TI, the establishment of an ambidextrous identity not only helped address immediate competitive pressures but also positioned the organisation for enduring resilience in the face of ongoing industry changes. As Langlois and Steinmueller (2000) argue, "the American

resurgence in the semiconductor industry can be attributed to three key factors. The second, organizational innovation and specialization, enabled American firms to leverage their structural advantages while capitalizing on global manufacturing capabilities." The shift in organisational structure provided Texas Instruments with the capabilities to compete with the highly specialised Asian semiconductor companies.

## Recommendations

---

Structural changes at Texas Instruments in the 1980s were met with resistance from some managers, as the new decentralised framework represented a significant departure from the organisational practices they had adhered to for over a decade. Resistance to change often stems from uncertainty, perceived threats to power dynamics, or attachment to familiar routines (Eggins et al., 2010). At TI, this resistance highlighted the challenges of introducing transformation in a well-established organisation.

According to Rosson (1988), strong leadership is key to overcoming resistance. Effective leaders must provide clarity around the vision for change and support their teams in adapting to new roles and processes. This was exemplified at TI, where a deliberate focus on management training and strategic communication helped employees understand the rationale behind the structural adjustments and align with the organisation's evolving goals and new identity. The management at TI provided extensive support to their employees throughout the transformation of the firm, promoting the new unified but independent, flexible identity of TI (Burrows, 1993; Ross, 1997).

As the organisation established a cohesive strategy and clear objectives under its new leadership, employees gradually adapted to the redefined identity of TI. This shift in organisational identity — towards being dynamic, innovative and ambidextrous — became a source of resilience. Research by Dick et al. (2018) emphasises that a flexible and adaptive organisational identity not only mitigates resistance during transformation but also prepares the organisation to respond effectively to future uncertainties.

There has been increased emphasis on the importance of harnessing social capital by organisations (Haslam et al., 2003). The ASPIRe model provides a structured approach to leveraging subgroup identities within the organisation to foster a stronger sense of organisational identification among employees. The model progresses through distinct phases. Initially, it identifies the social identities that employees use to define themselves (AIRing). This is followed by intermediate stages where subgroups, and then the broader organisation, align goals with these identities (Sub-Casing and Super-Casing). In the final phase (ORGanizing), organisational planning and goal setting are guided by the insights gained from the earlier stages. After this process, employee satisfaction and commitment to the outcomes, as well as organisational performance against the emergent goals, are evaluated, and these results serve as a foundation for refining and iterating the ASPIRe process in future (Haslam et al., 2003).

Managers at Texas Instruments can implement this model to manage the risks of transformation and the resistance to change coming from the business. The ASPIRe model works by assisting in the realignment of the organisational identity with the individual identities of employees — creating a shared identity that everyone has contributed to, rather than one imposed from the top down.

## Conclusion

Texas Instruments' journey through the 1980s highlights the critical importance of dynamic capabilities in navigating volatile and competitive environments. Faced with significant challenges, including the aggressive entry of Japanese competitors and internal cognitive inertia, TI demonstrated that transformation demands structural adaptability, identity shifts, and leadership committed to fostering innovation and resilience.

The segment data makes the effect visible: the Components segment's revenue grew from \$2 billion to \$9 billion in the space of ten years, and the share price rose from roughly \$1.00 in 1985 to around \$9.00 by 1996.

### Industry segment net revenues (millions of dollars)

| Segment                                | 1996           | 1995            | 1994           |
|----------------------------------------|----------------|-----------------|----------------|
| Components — trade                     | \$8,008        | \$9,419         | \$6,787        |
| Components — intersegment              | 63             | 60              | 56             |
| <b>Components total</b>                | <b>8,071</b>   | <b>9,479</b>    | <b>6,843</b>   |
| Digital Products — trade               | 1,717          | 1,829           | 1,661          |
| Digital Products — intersegment        | 7              | 23              | 1              |
| <b>Digital Products total</b>          | <b>1,724</b>   | <b>1,852</b>    | <b>1,662</b>   |
| Metallurgical Materials — trade        | 172            | 160             | 152            |
| Metallurgical Materials — intersegment | 12             | 23              | 25             |
| <b>Metallurgical Materials total</b>   | <b>184</b>     | <b>183</b>      | <b>177</b>     |
| Eliminations and other                 | (39)           | (105)           | (74)           |
| <b>Total</b>                           | <b>\$9,940</b> | <b>\$11,409</b> | <b>\$8,608</b> |

*Source: Texas Instruments annual reports. Note that Jerry Junkins passed away unexpectedly in 1996, making the financial data for that year something of an outlier.*

Under Jerry Junkins' leadership, TI's transition to a decentralised structure facilitated improved adaptability, enabled contextual ambidexterity, and countered cognitive inertia. This structural transformation not only mitigated resistance to change but also fostered an organisational identity centred on adaptability and dynamism, essential for sustained competitiveness in the semiconductor industry (Langlois & Steinmueller, 2000).

By embracing a flexible and innovative organisational identity, supported by strong leadership and clear strategic goals, TI positioned itself as a resilient force in a rapidly evolving market. The case demonstrates that organisations which invest in dynamic capabilities and prioritise transformation are better equipped to thrive in uncertain and competitive environments, providing a valuable blueprint for other firms facing similar challenges.

## References

---

- Anon. (1986). *Texas Instruments 1986 Annual Report*. Austin: Mergent Archives.
- Anon. (1987). *Texas Instruments 1987 Annual Report*. Austin: Mergent Archives.
- Anon. (1988). *Texas Instruments 1988 Annual Report*. Austin: Mergent Archives.
- Anon. (1989). *Texas Instruments 1989 Annual Report*. Austin: Mergent Archives.
- Anon. (1990). *Texas Instruments 1990 Annual Report*. Austin: Mergent Archives.
- Anon. (1991). *Texas Instruments 1991 Annual Report*. Austin: Mergent Archives.
- Anon. (1995). *Texas Instruments 1995 Annual Report*. Austin: Mergent Archives.
- Anon. (1996). *Texas Instruments 1996 Annual Report*. Austin: Mergent Archives.
- Anon. (2024). *Texas Instruments — 52 Year Stock Price History*. Available at: <https://www.macrotrends.net/stocks/charts/TXN/texas-instruments/stock-price-history> [Accessed 13 December 2024].
- Blanchette, I. & Dunbar, K. (2000). How analogies are generated: the roles of structural and superficial similarity. *Memory & Cognition*, 28(1), pp. 108–124.
- Borrus, M. G. (1988). *Competing for Control: America's Stake in Microelectronics*. Cambridge: Ballinger.
- Burrows, P. (1993). *TI is Moving Up in the World*. Available at: <https://www.bloomberg.com/news/articles/1993-08-01/ti-is-moving-up-in-the-world> [Accessed 23 November 2024].
- Dick, V. R., Ciampa, V. & Liang, S. (2018). Shared identity in organizational stress and change. *Current Opinion in Psychology*, 23, pp. 20–25.
- Eggers, R. E., Haslam, A. S. & Reynolds, K. J. (2010). The ASPIRe model: Actualizing Social and Personal Identity Resources to enhance organizational outcomes. *Journal of Occupational and Organizational Psychology*, 76(1), pp. 1–147.
- Haleblian, J. & Finkelstein, S. (1993). Top management team size, CEO dominance, and firm performance: The moderating roles of environmental turbulence and discretion. *Academy of Management Journal*, 36(4), pp. 844–863.

- Hambrick, D. C. & Mason, P. A. (1984). Upper Echelons: The Organization as a Reflection of Its Top Managers. *The Academy of Management Review*, 9(2), pp. 193–206.
- Haslam, S. A., Egghins, R. A. & Reynolds, K. J. (2003). The ASPIRe model: Actualizing Social and Personal Identity Resources to enhance organizational outcomes. *Journal of Occupational and Organizational Psychology*, 76(1), pp. 83–113.
- Hayes, T. C. (1987). *Texas Instruments vs. Japan*. Available at: <https://www.nytimes.com/1987/07/01/business/texas-instruments-vs-japan.html> [Accessed 28 November 2024].
- Hoffman, M., Lubell, M. & Hillis, V. (2014). Linking knowledge and action through mental models of sustainable agriculture. *Proceedings of the National Academy of Sciences*, 111(36), pp. 13016–13021.
- Langlois, R. N. & Steinmueller, W. E. (2000). Strategy and Circumstance: The Response of American Firms to Japanese Competition in Semiconductors, 1980–1995. *Strategic Management Journal*, 21, pp. 1163–1173.
- Miller, D. & Chen, M. J. (1994). Sources and consequences of competitive inertia: A study of the US airline industry. *Administrative Science Quarterly*, 39(1), pp. 1–24.
- Mintzberg, H. (1979). *The Structuring of Organisations*.
- Neck, C. B. & Neck, C. P. (2022). Seize the day: executive thought self-leadership and heterogeneity among dynamic managerial capability underpinning cognitive capabilities. *Journal of Management and Strategy*, 13(2).
- Ossenbrink, J., Hoppman, J. & Hoffmann, V. H. (2019). Hybrid Ambidexterity: How the Environment Shapes Incumbents' Use of Structural and Contextual Approaches. *Organization Science*, 30(6), pp. 1319–1348.
- Pederson, J. P. (2017). *International Directory of Company Histories*. 1st ed. Framington Hills: St. James Press.
- Pollack, A. (1982). *Japan's Big Lead in Memory Chips*. Available at: <https://www.nytimes.com/1982/02/28/business/japan-s-big-lead-in-memory-chips.html> [Accessed 12 December 2024].
- Ross, J. W. (1997). Texas Instruments Incorporated: Service Level Agreements and Cultural Change. *MIT Sloan*, September.
- Rosson, R. D. (1988). Self-Directed Work Teams at Texas Instruments Defense Systems & Electronics Group. *MIT Sloan School of Management*.
- Schuyten, P. J. (1979). *Winning and Losing at Texas Instruments*. Available at: <https://www.nytimes.com/1979/05/13/archives/winning-and-losing-at-texas-instruments-how-competitors-move-in-on.html> [Accessed 13 December 2024].

Smolinski, S. M., Sieweke, J. & Zapkau, F. B. (2020). The Relationship Between Top Management Team Diversity and Firm Innovation Output: A Meta-Analysis. *Academy of Management*, 2020(1).

Teece, D. J. (2007). Explicating Dynamic Capabilities: The Nature and Microfoundations of (Sustainable) Enterprise Performance. *Strategic Management Journal*, 28, pp. 1319–1350.

Teece, D. & Pisano, G. (1998). The dynamic capabilities of firms: An Introduction. In: *Technology, Organisation, and Competitiveness: Perspectives on Industrial and Corporate Change*. Oxford Academic, pp. 193–212.